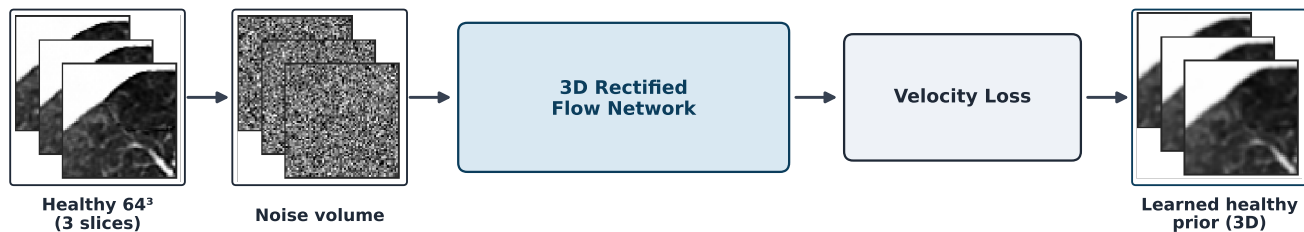


Training (offline)



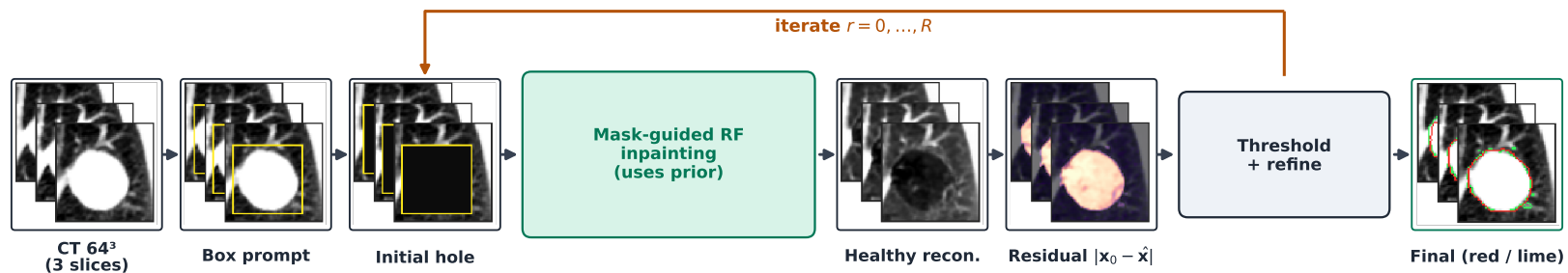
$$\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$\mathbf{x}_t = (1 - t/T)\mathbf{x}_0 + (t/T)\epsilon$$

$$\mathcal{L}(\theta) = \mathbb{E}[\|\mathbf{v}_\theta(\mathbf{x}_t, t) - \mathbf{u}\|_1]$$

→ inference

Inference (online)



$$\rightarrow \mathbf{h}^{(0)}$$

$$\mathbf{h}_z^{(0)} = \text{AABB}_{2D}(\mathbf{y}_z) \odot \rho$$

$$\mathbf{x}_t \leftarrow \mathbf{m} \odot \tilde{\mathbf{x}}_t(\mathbf{x}_0) + (1 - \mathbf{m}) \odot \mathbf{x}_t$$

$$\mathbf{x}_t \leftarrow \mathbf{x}_t + \mathbf{v}_\theta(\mathbf{x}_t, t) \Delta t$$

$$\hat{\mathbf{x}}^{(r)}$$

$$\mathbf{d}^{(r)} = |\mathbf{x}_0 - \hat{\mathbf{x}}^{(r)}|$$

$$\mathbf{h}^{(r+1)} = \mathbf{1}[\mathbf{d}^{(r)} > \tau] \odot \text{Dilate}(\mathbf{h}^{(r)}; \rho)$$

$$\hat{\mathbf{y}} = \mathbf{h}^{(R)}$$

Mask refinement

$$\mathbf{h}^{(r+1)} = \mathbf{1}[\mathbf{d}^{(r)} > \tau] \odot \text{Dilate}(\mathbf{h}^{(r)}; \rho), \quad \hat{\mathbf{y}} = \mathbf{h}^{(R)}$$

